Gao 2021 Breast — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

5/6 breast samples (≥347 cells) intermix on one continuous manifold with no per-patient islands (radius-graph: 89 detached removed; kNN = 1 component); both compartments (Embryonic 5,242 / Cycling 1,080) carry the trajectory.

Gate 2 — module geography → GO (non-branching)

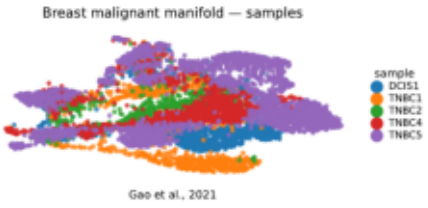
Organised Gast↔pSom cross-repression (r=−0.49) laid out along one axis; Crest anti-correlated with pSom (r=−0.58), rising to a terminal readout (ρ=+0.23; crest-high decile mean pt=0.75). Along pseudotime Gast declines (ρ=−0.31); no Gast/pSom co-expression hub (395 vs 731 exp.).

Gate 3 — hub mutual exclusivity → NA

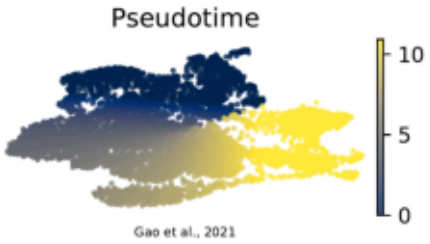
Single cross-repression axis — no bifurcation and no Gast/pSom co-expression hub, so there is no branch pair to test for mutual exclusivity.

STEP FIGURES

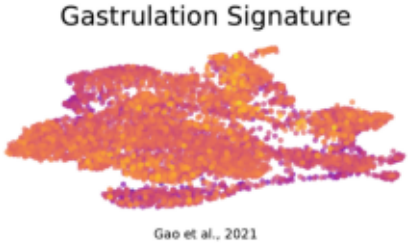
Gate 1: integration



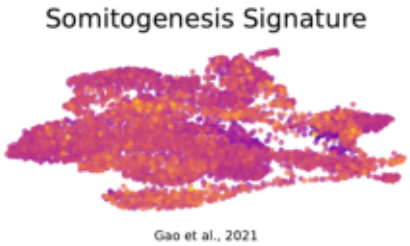
Trajectory / root



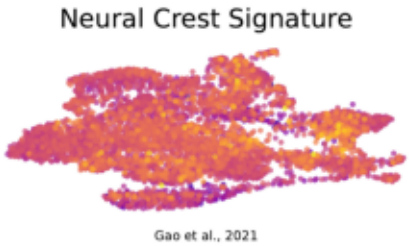
Gast



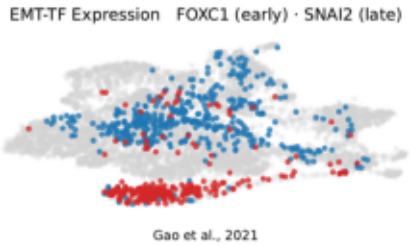
pSom



Crest



EMT-TFs



Dataset. Breast Cancer (Gao et al., 2021), 10x — 11 patients / 29,992 cells (master scope, incl. thyroid). Breast-only malignant: 6,606 pooled across both matrix leaves (5 anaplastic-thyroid samples excluded), 6,411 across 5 samples (≥347) after threshold, 6,322 after island removal.

Verdict. ✓ Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. Gao's breast malignant manifold is a single axis of organised Gast↔pSom cross-repression (r=−0.49) with a terminal neural-crest gradient (non-branching, Linear-pSom/Crest); it lacks Puram's hub-and-spoke bifurcation (no co-expression hub), so Gate 3 has no branch pair to test. The proliferation split is orthogonal to the developmental axis (root tercile 4.5% cycling vs terminus 40%), so the whole manifold is kept (Bischoff/Sun/Lee no-scoping precedent). EMT-TFs FOXC1 (early/root) → SNAI2 (late/terminus); ZEB1/TWIST2 near the detection floor (1.1%/0.3%); end-separation weak (Δpt=0.25).

Parameter choices

Min cells/sample	≥347 (gap 618→195)	**Integration**	Harmony θ=10; PCA50; spread 1.3
Scope	whole manifold; breast-only	**Trajectory**	scFates curve N=20 λ=0.3 μ=100
Root node	tip 2 = Gast/pSom-high	**Compartments**	Embryonic 5,242 / Cycling 1,080